

Two ways to be a value

Algebraic data types and objects

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The question

What is a datatype? What is an object? What is an algebraic data type?

What is a datatype?

Some answers we would all give:

- `int` — a set of numbers?
- `int32` — the same, but exactly 2^{32} of them?
- `struct` — several values glued together?
- `class` — values with code attached?
- “a collection of values” — of what, arranged how?

Each describes one aspect. The question returns in four later sessions:

S3 • today	how a datatype organises its values
------------	--

S8	a type as a protocol
----	-----------------------------

S9	what a type claims about a value
----	---

S11	what a type hides
-----	--------------------------

S12	a type as a relation
-----	-----------------------------

What is an object?

In everyday use, “object” bundles several design decisions. Five of them, each treated on its own day:

bundled into “object”	actually a separate axis	
mutable fields, identity	state vs. persistence	S2, S7
inheritance, extends	subtyping	S10
classes	classes vs. prototypes (Self, JS)	today, one slide
declared implements	nominal vs. structural typing (Java vs. Go)	S10
static types	Smalltalk has none	S9

In this session, an object is a value used only through the operations it carries.

State and identity can be added — and usually are — but they are not essential.

What is an algebraic data type?

Last week we read two of them without the name — and used the first in the type of `pop`:

```
type 'a option = None | Some of 'a
type 'a list   = []    | (::) of 'a * 'a list
```

```
val pop : 'a t -> ('a * 'a t) option
(* the value and the new version – or nothing *)
```

Every value is built from a **closed list of shapes**, and taken apart the same way.

By the end of this first hour: why a closed list of shapes is the **natural design** for many datatypes — and what the opposite design looks like.

Two ways to be a set

The discovery, in JavaScript

One interface

A set of integers. Four operations, and nothing else:

```
empty           // the set with no elements  
insert(s, n)    // the set s, with n added  
contains(s, x)  // is x in s?  
union(s, t)     // every element of s, or of t
```

Client code uses only these four names.

Two implementations follow. The client cannot tell them apart — until the interface grows.

Way 1 — a set **is data**

```
const empty1      = [];  
const insert1     = (s, n) => s.includes(n) ? s : [...s, n];  
const contains1   = (s, x) => s.includes(x);  
const union1      = (s, t) => [...s, ...t.filter(x => !s.includes(x))];
```

Every operation **inspects*** the array.

`union1` inspects **both** of its arguments.

*inspect: read the representation directly — here as an array; in OCaml, as a constructor and its arguments.

Way 2 — a set **is a function**

A set is a function that answers one question: *is x inside?*

```
const empty2    = x => false;
const insert2   = (s, n) => (x => x === n || s(x));
const contains2 = (s, x) => s(x);
const union2    = (s, t) => (x => s(x) || t(x));
```

Every operation **calls**^{*} the set. `contains2` is just the call.

`union2` can do nothing with `s` and `t` except call them. **There is no inside.**

^{*}call: use a value only through the interface it exposes, learning nothing about how it is made.

Two inserts, unfolded

Way 1

```
const s3  = insert1(empty1, 3)
      // = [3]
const s35 = insert1(s3, 5)
      // = [3, 5]
```

Each step produces **data**: an array anyone may inspect.

Way 2

```
const s3  = insert2(empty2, 3)
      // = x => x === 3 || empty2(x)
const s35 = insert2(s3, 5)
      // = y => y === 5 || s3(y)
```

Each step produces a **closure**^{*}. s35 holds s3; nothing runs until someone calls it.

Both s35 are chains of length two: [3, 5], and a function holding 5 and s3, which holds 3 and empty2. Same shape — and, from the operations' side, one can be inspected while the other can only be called.

^{*} closure: a function together with the bindings it captured where it was defined — here n and the previous set.

Same client, same answers

```
const a1 = insert1(insert1(empty1, 3), 5), b1 = insert1(empty1, 4);  
const a2 = insert2(insert2(empty2, 3), 5), b2 = insert2(empty2, 4);  
const test1 = [3, 4, 5, 6].map(x => contains1(union1(a1, b1), x));  
const test2 = [3, 4, 5, 6].map(x => contains2(union2(a2, b2), x));
```

Run

```
way 1: [true, true, true, false]  
way 2: [true, true, true, false]
```

Nothing so far tells the two apart. But what if we **extend** the interface with

1. a new operation
2. an operation on two sets
3. a new kind of set

Extension 1 — a new operation: isEmpty

Way 1

```
const isEmpty1 = s => s.length === 0;
```

One line. Nothing else is touched.

```
const e1 = [empty1, insert1(empty1, 3),  
            union1(empty1, empty1)].map(isEmpty1);
```

Run

```
e1 = [true, false, true]
```

Way 2

```
const isEmpty2 = s => ???
```

A function cannot be asked whether it ever answers true — unless a set carries **two** functions:

```
const empty2b = { contains: x => false,  
                  isEmpty: () => true };  
const insert2b = (s, n) => ({  
  contains: x => x === n || s.contains(x),  
  isEmpty: () => false });  
const union2b = (s, t) => ({  
  contains: x => s.contains(x) || t.contains(x),  
  isEmpty: () => s.isEmpty() && t.isEmpty() });
```

So a set becomes a **record of operations** sharing hidden data — and every maker of sets, empty, insert and union, is rebuilt this way.

Extension 2 — an operation on two sets

Way 1

```
const union1 = (s, t) =>
  [...s, ...t.filter(x => !s.includes(x))];

const equal1 = (s, t) =>
  s.length === t.length &&
  s.every(x => t.includes(x));
```

Both inspect both arrays: merge them, or compare them.

Way 2

```
const union2 = (s, t) => x => s(x) || t(x);

const equal2 = (s, t) => ???
```

Both can only call. Enough for `union` —

not enough for `equal`: no finite number of calls decides whether two functions agree on every input.

Way 1: an operation on two sets sees both representations — it can merge them, or compare them.

Way 2: it can only ask each set about an `x` and combine the answers. The two sets themselves are **never compared**.

Extension 3 — a new kind of set: the even numbers

Way 1

No array holds the even numbers. There are now **two kinds** of set, told apart by a tag:

```
{ kind: 'finite', xs: [3, 5] } // as before
{ kind: 'evens' }             // no array
```

```
const contains1 = (s, x) =>
  s.kind === 'evens' ? x % 2 === 0
    : s.xs.includes(x);

const isEmpty1 = s =>
  s.kind === 'evens' ? false
    : s.xs.length === 0;
```

— and **every** operation now begins by **reading the tag**. `equal1` gains one case per pair of kinds.

Way 2

```
const evens = { contains: x => x % 2 === 0,
                  isEmpty: () => false };
const a2b = insert2b(insert2b(empty2b, 3), 5);
const test3 = [3, 4, 5, 6].map(x =>
  union2b(a2b, evens).contains(x));
```

One record, and it holds *every* even number. The other makers and every client stay as they are.

Run

```
test3 = [true, true, true, true]
```

Extension 3, continued — the union of two kinds

Way 1

$\{3, 5\} \cup \text{evens}$ is **neither kind**. The operation keeps what it cannot compute, as a third kind:

```
const union1 = (s, t) =>
  s.kind === 'finite' && t.kind === 'finite'
    ? finite(merge(s.xs, t.xs))
    : { kind: 'union', s, t };
```

Run

```
test4 = [true, true, true, true]
```

In Way 1 every result must be one of the kinds. A new kind must keep **every operation closed** — or it costs one more kind, and one more case in every operation.

Every operation gains a case. In `contains1`:

```
s.kind === 'union' ? contains1(s.s, x)
                  || contains1(s.t, x)
```

— `union2`, written as `data.isEmpty1` asks both halves too. `equal1` has no case to write: two unions cannot be compared without a normal form.

Way 2

Nothing to do. `union2b` already returns a set: a record of two functions is always a valid set.

Two ways, stated

	Way 1 — a set is data	Way 2 — a set is a function
a new operation	one new function	every maker rebuilt
an operation on two sets	inspect both	call both — no comparison
a new kind of set	every operation changed	one new record

Way 1: a value **is** one of a closed collection of representations, open to every operation. A new operation is free; a new representation affects every operation.

Way 2: a value **is** its own operations, bundled together when it is made. A new implementation is free; a new operation affects every implementation.

Two ways of **implementing one interface**. Neither is a language feature — we used arrays, arrow functions and records. The names come next.

Algebraic data types

Way 1, made precise

Naming the two ways

what we built

Way 1 — a value is data; operations inspect it

Way 2 — a value is a function; operations call it

what it is called

algebraic data types

objects

Both are older than the languages we write them in. Reynolds (1975) called them *user-defined types* and *procedural data structures* — “**complementary approaches to data abstraction**”.

Two ADTs, one acronym.

Abstract data type (Liskov & Zilles, 1974): a type its clients know only through its operations — session 11.

Algebraic data type (Hope, 1980): a type whose values come in a closed list of shapes — today.

An enumeration

```
type note = C | D | E | F | G | A | B
```

```
let semitones n = match n with  
  | C -> 0 | D -> 2 | E -> 4 | F -> 5 | G -> 7 | A -> 9 | B -> 11
```

Reading rule: | is **or**. A note is C, or D, or ...

The constructors are values we **invent**. C is not 0; nothing is inside it. It is a **tag**, as kind: 'evens' was for the set — but one of exactly seven values of a new type, *note*, each distinct from the others by construction.

The usual encoding in C, and in Java before `enum`, was `static final int C = 0, D = 2, ...`
Now `C + D` compiles, and 7 is accepted wherever a note is expected. A variant type allows neither.

Carrying values: products and sums

```
type pitch      = { note : note; octave : int }  
type duration = Whole | Half | Quarter | Eighth  
type symbol    = Note of duration * pitch | Rest of duration
```

- `pitch` is a **product** type: each of its values holds every field.
- `symbol` is a **sum** type: each of its values is exactly one alternative — a `Note` carrying a duration and a pitch, or a `Rest` carrying a duration. That is what `of` reads as.

A `Rest` carries no pitch: no field is left empty or set to `null` — a rest with a pitch cannot be written at all.

```
let beats s = match s with  
  | Note (d, _) | Rest d -> beats_of d
```

Why “algebraic”

Count the values of a type

<code>unit</code>	1
<code>bool</code>	2
<code>note</code>	7
<code>duration</code>	4
<code>pitch = note × int</code>	7 · int
<code>symbol = duration × pitch + duration</code>	4 · 7 · int + 4

The operations on types, $\times$ and $+$, **form an algebra**. Counting is its simplest model — and it works for infinite types too, as laws rather than numbers.

Products multiply, sums add

$$|A \times B| = |A| \cdot |B| \quad |A + B| = |A| + |B|$$

The laws of $+$ and $\times$ hold as **isomorphisms** of types, with a conversion function each way:

$$A \times (B + C) \cong A \times B + A \times C$$

Two encodings of one type

A struct with a tag

```
struct symbol {  
    int tag;          /* 0 = note, 1 = rest */  
    int duration, note, octave;  
};
```

2^{128} states, four 32-bit ints. Most mean nothing: a rest with a pitch; a tag of 7.

A sum

```
type symbol =  
    | Note of duration * pitch  
    | Rest of duration  
    (* two shapes, nothing else *)
```

$4 \cdot 7 \cdot |\text{int}| + 4$ states. Each means something.

Next to a struct that holds every field, a sum **removes the meaningless states**.

option absence as **one explicit case**: None | Some of 'a has one more value than 'a, and every match must handle it.

null absence added to **every reference type at once** and recorded in none: no match is required, but the missing case surfaces at runtime as a NullPointerException.

Recursion: a type mentioned in its own definition

```
type 'a list = [] | (::) of 'a * 'a list
```

```
type set = Empty | Insert of int * set      (* slide 6, as a type *)
```

```
let rec contains s x = match s with  
  | Empty          -> false  
  | Insert (n, rest) -> n = x || contains rest x
```

Its equation, solved: $L = 1 + aL \Rightarrow L = \frac{1}{1-a} = 1 + a + a^2 + a^3 + \dots$

A list is empty, or one element, or two, or three ... — **a tuple of some length**, one shape for each length.

The set we wrote as an array on slide 6 is now a type. The closed list of shapes — Empty, Insert — is written down **where the compiler can read it**.

Constructors* are functions — but not ordinary ones

```
length : 'a list -> int           (* an ordinary function *)
```

```
Note   : duration * pitch -> symbol (* a constructor *)
```

An ordinary function may **lose information**: from `length xs` we cannot recover `xs`; many lists share one `length`.

A constructor **loses no information** — three guarantees no function comes with:

injective	different carried values give different values
------------------	--

disjoint	different constructors never build the same value
-----------------	---

exhaustive	every value of the type came from some constructor
-------------------	--

So every value was built in **exactly one way**: one constructor, applied to the values it carries.

* Unlike Java's constructors, these run no code and can be taken apart: a tag with a payload.

Matching recovers the carried values

```
let s = Note (d, p)           (* forwards: build *)
match s with Note (d, p) -> ... (* backwards: recover d and p *)
```

A constructor goes from the carried values to the value: $(d, p) \mapsto \text{Note } (d, p)$.

A match goes **from the value back to the carried values**: $\text{Note } (d, p) \mapsto (d, p)$.

It can, because every value was built in exactly one way.

- We cannot match on $f \ x$ for an arbitrary f : there is no inverse to run.
- We cannot match on a Way 2 value: any function may have produced it, and nothing in it says which, or from what — its environment is sealed, and there is no fixed list of makers.
There is nothing to invert.
- The list of shapes used to live in a comment. Now it lives in the type, and **the compiler reads it**: every match covers every shape, and no case is unreachable.

Algebraic data types: a timeline

1970s	Pascal variant records; C <code>union</code> with a tag field — the list of shapes is a convention, unchecked
1977	NPL (Burstall): types built from constructors, and patterns on them
1980	Hope (Burstall, MacQueen, Sannella): the closed declaration — type and constructors together
1984 →	Standard ML, then OCaml and Haskell: the ML family makes them the default
2014 →	Swift enums, Rust enums, Kotlin sealed classes, TypeScript discriminated unions
2021 →	Python 3.10 <code>match</code> ; Java 17 <code>sealed</code> , Java 21 pattern <code>switch</code>

“Algebraic” also names the mathematics: the values of the type are **exactly the terms** built from its constructors, nothing more and nothing identified — the *initial algebra* of its constructors (Goguen, Thatcher, Wagner, Wright, 1977).

Objects

Way 2, made precise

The set as objects

```
type set =  
  < contains : int -> bool; is_empty : bool >  
  
let empty : set = object  
  method contains _ = false  
  method is_empty = true  
end  
  
let insert (s : set) n : set = object  
  method contains x = x = n || s#contains x  
  method is_empty = false  
end
```

type set is **the interface** of a set — extension 1's record of operations, as a type — and under it two kinds of set, empty and insert^{*}.

In the Way 1 type they were the constructors Empty and Insert; here the type lists only **operations**, and the kinds live in the code.

No match on a set: nothing at compile time says how many kinds exist, or which one we hold.
That is the point: which kind we hold is decided at run time, by the object, when it is called — **dynamic dispatch**.

^{*} object ... end builds one object and # calls a method. No class is involved. All four kinds: demos/set_way2.ml.

The same, in Java

```
abstract class Set {
    abstract boolean contains(int x);
    abstract boolean isEmpty();
}
class Empty extends Set {
    boolean contains(int x) { return false; }
    boolean isEmpty()      { return true; }
}
class Insert extends Set {
    private final Set s; private final int n;
    Insert(Set s, int n)
    { this.s = s; this.n = n; }
    boolean contains(int x)
    { return x == n || s.contains(x); }
    boolean isEmpty()      { return false; }
}
```

The abstract class is the interface^{*}; each subclass is one kind, with its own operations.

The fields are private^{**}: from a client's side, a Set is only what its methods answer.

^{*} A Java interface says the same, without instance fields. All four kinds: `demos/SetWay2.java`.

^{**} `private`: no client reads the field. `final`: assigned once, in the constructor — a set never changes after it is made.

A new kind, in Java

A new kind is one more class:

```
class Evens extends Set {  
    boolean contains(int x) { return x % 2 == 0; }  
    boolean isEmpty()      { return false; }  
}
```

Set, Empty, Insert and every client **stay as they are**:

```
Set s = new Insert(new Insert(new Empty(), 3), 5);  
s.contains(4);
```

Which contains runs is decided by the object `s` holds, at run time — three calls down the chain, none visible to the caller.

Classes

Nothing so far needed a class. In everyday languages, a class bundles several jobs:

- **factory** — new makes objects
- **code sharing** — methods written once
- **type** — Set in signatures
- **privacy boundary**
- **inheritance** — session 10
- **namespace** — Set.empty

Languages disagree on which to bundle, and on whether to have a class at all:

Java

Smalltalk

JavaScript

OCaml

Go

```
public boolean equals(Object o) {  
    return o instanceof Insert i && i.n == n && i.s.equals(s);  
}
```

All six jobs in one box. Privacy is **per class**, not per object: equals reads another object's n. The static methods of a class are the operations of an abstract data type.

Abstract data types versus objects

What each hides, and how each grows — Reynolds 1975 · Cook 2009 · Wadler 1998

One abstraction barrier, placed two ways

Both ways hide how an operation works: a client calls `contains` and gets an answer, never seeing the body — the array scan, or the chain of calls. Any function does this much. What each way adds differs:

[ADT] Way 1, behind a wall we will build in session 11. The **representation type** is hidden: behind an `.mli`, as session 2's stack, `type t` shows no constructors, so no client can match on a set or build one by hand. Cook calls this **type abstraction**. Only a compiler can enforce it.

[Objects] Way 2, as it stands. Nothing is hidden at the type level: the interface type is public. What is hidden is the environment of one closure — this `s`, this `n` — and it is hidden by being a function: nothing but a call can reach in. Cook calls this **procedural abstraction**: a procedure's wall, with the data inside.

One wall around the whole type, or **one wall around each value**.
Objects need no type system to keep their secrets; abstract data types do.

Complementary approaches to data abstraction

	abstract data type	object
what is abstracted	the representation type	the procedures in a value
operation bodies	hidden	hidden
representation	one, hidden	none in the interface
an operation is written	once, for every value	by each value, for itself
sees another value's inside	yes — <code>union1</code> , <code>equal1</code>	no — <code>union2</code> ; no <code>equal2</code>
implementations	one per type; two do not mix	any number, one type; they mix

Reynolds (1975) saw the two as **complementary**; Cook (2009) gave the table its rows.

Neither is a weaker form of the other. An object is not an abstract data type missing its type: it hides its data another way. An abstract data type is not an object missing dispatch: every value of `t` runs the same `contains`, and its `match` chooses the kind — the value does not.

They hide different things, by different means — the compiler, or the closure.

The matrix: kinds × operations

	contains	isEmpty	union	equal
Empty	✓	✓	✓	—
Insert	✓	✓	✓	—
Evens	✓	✓	✓	—
Union	✓	✓	✓	?

Algebraic data types (Way 1): one function per operation, every kind inside it — a **column** at a time.

One more operation is one more column, `equal` included — except on a `Union`, never computed.

Objects (Way 2): one class per kind, every operation inside it — a **row** at a time.

One more kind is one more row, `Union` included. `equal` gets no row: a row cannot inspect another.

Wadler (1998) named the problem of growing both ways at once, with the compiler still checking every cell: **the expression problem**. Its resolutions are session 10.

One language, both ways

Way 2 — abstract class and subclasses

```
abstract class Set { ... }  
class Empty extends Set { ... }  
class Insert extends Set { ... }  
class Evens extends Set { ... }  
class Union extends Set { ... }
```

Rows. The part before, as written.

Way 1 — sealed, records, switch

```
sealed interface Shape  
  permits Circle, Square, Tri {}  
record Circle(double r) implements Shape {}  
record Square(double s) implements Shape {}  
record Tri(double b, double h) implements Shape {}  
  
static double area(Shape sh) {  
  return switch (sh) {  
    case Circle c -> Math.PI * c.r() * c.r();  
    case Square s -> s.s() * s.s();  
  };  
}
```

permits is |, record is of, switch is match. Java required a quarter of a century, and three additions to the language, before it could express the second form directly, with the compiler checking the cases. That delay is the measure of its **default**.

Exhaustiveness in three languages

One case left out of a match, on purpose. Three compilers, three answers:

OCaml

```
Warning 8 [partial-match]: this pattern-matching is not exhaustive.  
Here is an example of a case that is not matched: Repeat (_, _)
```

Java 21+

```
A.java:10: error: the switch expression does not cover all possible input values
```

Python 3.10+

```
square: 9  
triangle: None
```

OCaml warns and **names a missing value**. Java refuses to compile. Python runs, and the missing case becomes a None that travels on.

OCaml: the lab's melody with Repeat added and no match updated. Java, Python: Tri left out of area.

The checks are dual

Way 1 — by column

```
Warning 8 [partial-match]:  
this pattern-matching is not exhaustive.  
Here is an example of a case that  
is not matched: Repeat (_, _)
```

Does every operation handle every kind?

Way 2 — by row

```
B.java:10: error: Evens is not abstract  
and does not override abstract method  
isEmpty() in Set
```

Does every kind provide every operation?

Each check is blind where the other sees. Extension is blocked exactly where the **declaration site belongs to someone else**: the permits list, or the abstract class.

An instanceof chain is not a match: nothing checks it — hence the visitor.

Visitor: pattern matching by hand

```
interface Visitor<R> {
    R onEmpty();
    R onInsert(Set s, int n);
}
abstract class Set {
    abstract <R> R accept(Visitor<R> v);
}
class Empty extends Set {
    <R> R accept(Visitor<R> v)
    { return v.onEmpty(); }
}
class Insert extends Set {
    <R> R accept(Visitor<R> v)
    { return v.onInsert(s, n); }
}
```

Way 1 would write one match, an arm per kind. Java has none; we build it by hand:

- **the visitor**: one method per kind, `onEmpty` and `onInsert` — the arms
- **accept**: each kind's one new method; it calls its own arm and hands over what it carries
- `s.accept(v)` **dispatches twice**: the set picks `accept`, `accept` picks the arm. **The set chose the arm, not the caller.**

Way 1's trade, cost included: `isEmpty` is one visitor, written outside; a new kind needs `onEvens` in every visitor.

A visitor packages the arms of a match as methods, one per constructor.

What each way is good for

Objects

- **extension by strangers** — a framework defines `Widget`; plug-ins add kinds it never sees
- **state that changes, behind a wall** — a counter, a connection: only its own methods touch it (Parnas, 1972)
- **collections sharing only an interface** — a list of sets, of any kinds

Algebraic data types

- **closed domains, modelled exactly** — messages, UI states, `Option`
- **programs that process trees** — compilers, interpreters, query planners, JSON: few node kinds, many operations, some on two nodes at once
- **code that grows by a kind** — the compiler lists every site to fix
- **data that travels** — files, the network: only a tag and its payload can go

Much “object” code is data with getters — the shape that `record` and `sealed` were added to Java to fix.

Conclusion

Language defaults · the question in later sessions · the lab's type

Every language chooses a default

language	default	the other way
OCaml	Way 1	<code>object ... end</code> — rarely used
Rust, Swift, Haskell	Way 1	<code>dyn Trait</code> , protocols, type classes
Java	Way 2	<code>sealed + record + switch</code> , since 2021
Smalltalk	Way 2 only	no static list of kinds — session 9
Python, JavaScript	Way 2 , untyped	Python's <code>match</code> , unchecked; a <code>kind</code> field by hand
Go	Way 2 , by shape	a type <code>switch</code> , unchecked for missing cases

Paradigm $\neq$ language: a language only makes some choices the default. The friction of saying the other thing is the **evidence of the default**.

The same question in sessions 8, 9, 11 and 12

This session answered *what is a datatype?* by how it **organises** its values: by construction, or by behaviour. Four later sessions answer it again:

S8 a type as a **protocol**: which messages, in which order — session types

S9 what a type **claims** about a value: a type as a proof; Smalltalk's one big sum

S11 what a type **hides**: abstract data types. Way 1 behind a wall; session 2's `.mli` returns

S12 a type as a **relation**: tables, queries; a query plan is a tree — a Way 1 program

Lab 3: the melody type

```
type melody =  
  | Note of duration * pitch  
  | Rest of duration  
  | Seq of melody * melody    (* after *)  
  | Par of melody * melody    (* together *)
```

Every function in the lab is one match on this type. `melody.ml` has the holes; `dune exec ./main.exe` reports each one.

- | | |
|------|---|
| A | <code>count_notes</code> , <code>length_in_beats</code> : Seq adds, Par takes the maximum |
| B | <code>to_events</code> , then a <code>.wav</code> of Bach's Invention in C; add Repeat, follow the warnings |
| C | <code>transpose_octave</code> , <code>pretty</code> , <code>retrograde</code> , <code>simplify</code> |
| read | CS3110 chapter 3: sections 3.1, 3.5, 3.6, 3.8, 3.11, 3.12 |